



Aerodynamic efficiency and performance enhancement of fixed-wing UAVs

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ABSTRACT

The present work aims in providing guidelines for the aerodynamic and performance optimization of fixed-wing Unmanned Aerial Vehicles (UAVs) that operate in the low-speed subsonic regime. Using a combination of textbook methods, low-fidelity and high-fidelity aerodynamic analysis tools, an aerodynamics and drag breakdown analysis is made on a reference UAV wing platform. Then, a complete drag bookkeeping study is made on the corresponding reference UAV configuration, providing the contribution of each part to the total drag budget for various angles of attack. Using the results of the aerodynamic investigations as guide, a thorough drag-reduction technologies screening is made based on existing literature, with the discussion also including integration and realization issues. Concluding, the most promising technologies are proposed taking into account the potential benefits as well as the technology readiness. The current work can serve as a base for future, more detailed, aerodynamic and performance enhancement studies for UAVs.

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1. Introduction

Radio-controlled aircraft date back to the World War I, where they were mostly used as missiles or target drones [1]. However, it was not until the 1960s, that remote controlled aircraft started carrying more sophisticated equipment, for military reconnaissance purposes. Since then, the rapid advances in aeronautical industry, electrooptical systems, and mission planning algorithms, have allowed the design and large-scale production of Unmanned Aerial Vehicles (UAVs), capable of performing a wide variety of missions. At present, they are being operated by several military forces, whereas there is a growing need for unmanned platforms for civilian applications as well [2]. Indicatively, recent research [3] shows that “the global UAV market is projected to grow to USD 19.85 billion by 2021, at a compound annual growth rate (CAGR) of nearly 13% over the forecast period”.

In a very broad classification, UAVs can be divided to the multi-rotor and the fixed-wing ones, based on their fundamental lifting mechanism. Although being relatively cheap and maneuverable, the multi-rotors are used for entertainment and small-scale surveillance for the time being, due to their considerable limitations in payload capacity and flight endurance. Those limitations do not apply to the fixed-wing unmanned platforms and especially

the medium-sized tactical and Medium Altitude Long Endurance (MALE) ones, which, as a result, have a much broader mission range. Fixed-wing UAVs are ideal solutions for a wide range of operations, e.g. fire detection, search and rescue, coastline and sea-lane monitoring, as well as security surveillance and defense-related missions [4]. They feature several advantages, such as the low operating cost, the ability to operate under adverse or hazardous conditions, and the increased flight endurance, which is one of the most important characteristics when it comes to the aforementioned missions. In the case of an unmanned vehicle, where the endurance is only limited by the available fuel on-board, it is essentially up to the efficiency of the configuration to maximize the flight time, whereas, by enhancing the performance specifications (maximum speed, stall speed, rate of climb, turning radius etc.), the versatility and potential of fixed-wing UAVs can be further expanded. Hence, conducting research and developing dedicated guidelines, layouts and techniques to reduce the weight, optimize the aerodynamics, and improve the overall performance of fixed-wing UAVs is a very important step for the field of unmanned aviation.

There are several existing studies that describe the design of fixed-wing UAVs [5–13]. The corresponding aerial vehicles may vary in size, configuration type, energy source, etc. However, little to no effort is made on optimizing the aerodynamic efficiency and performance of those configurations and, even then, the analysis is relatively limited depending on the case, such as the winglet studies conducted in [7,14]. A very recent paper towards this direc-

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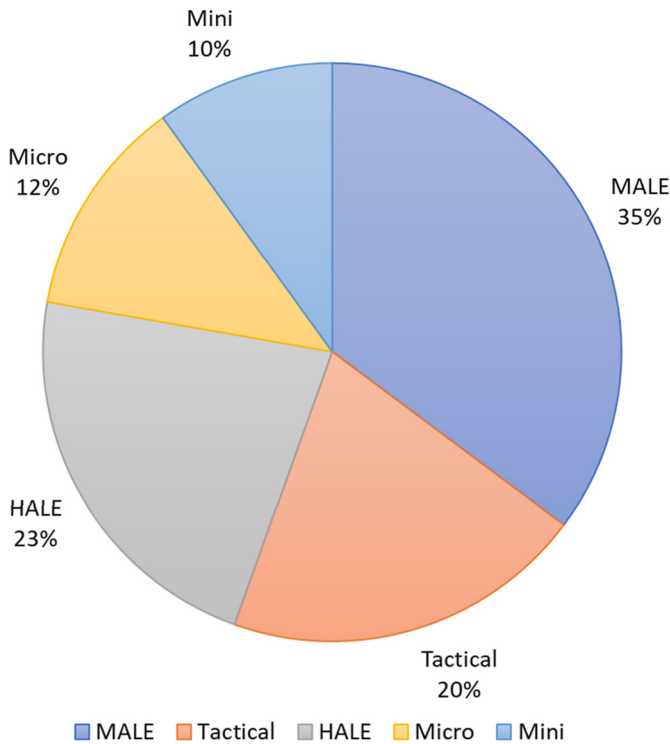


Fig. 1. Global Unmanned Aerial Vehicle (UAV) market share by UAV type (Data taken from [3]).

tion is the work of Goetten et al. [15], where a drag breakdown is made for four different UAV configurations. The aim of the paper is to compare different aerodynamic coefficient estimations methods, though, and no discussion is made on how to actually enhance the aerodynamic efficiency of the examined UAV configurations. That is, rather surprisingly, not enough information exists in the literature regarding the drag bookkeeping of fixed-wing UAVs, or the investigation of possible drag reduction techniques and layouts.

In an attempt to address this gap, the goal of the current paper is to investigate and provide guidelines for the aerodynamic and performance optimization of aerial vehicles that operate at low-subsonic, incompressible flow conditions and, more specifically, of fixed-wing UAVs. The current research emphasizes the tactical and MALE UAVs categories, since they are arguably the largest ones in terms of market share (Fig. 1). Compared to larger UAVs, they are

smaller, easier, and cheaper to manufacture, deploy and operate, while they are large enough to carry several kilograms of payload equipment, which may range from humanitarian aid to sophisticated electronics.

2. Methodology and tools

In this section a presentation of the methodology is made, followed by an overview of the theoretical background that includes a short listing of the main sources of drag. Then, a thorough description of the aerodynamic analysis tools that were employed during the current study is made, i.e. the textbook methods, the low- and the high-fidelity tools.

2.1. Methodology

Initially, a medium-size, fixed-wing reference UAV platform is defined. An aerodynamic analysis and drag breakdown are in turn conducted on the wing platform, followed by a drag bookkeeping study of the entire configuration. Meanwhile, a detailed drag-reduction technologies screening is made. The screening is based on a thorough literature review and includes both conventional (already in-service) and novel (not yet operational) technologies and techniques, which have been developed for aviation applications in general. Following this review, and also based on the results of the drag bookkeeping study, an initial drag-reduction technologies evaluation is made for UAVs and the appropriate conclusions are drawn. The roadmap of the current study is presented in Fig. 2.

As a start, the contribution of the current paper is twofold, i.e. to provide a typical UAV drag bookkeeping analysis (wing and integrated configuration) on the one hand, and an initial UAV aerodynamic efficiency and performance enhancing technologies listing on the other. Those results yield initial trends and guidelines, which are currently unavailable in the field of UAV design, for aerodynamic efficiency and performance enhancement studies. It should be noted, though, that this work is part of a larger-scale research, and that both the drag bookkeeping and the technology screening serve as a base for future, more detailed studies on the field. Having identified the main sources of drag for each component, and the key efficiency-enhancing technologies, a more targeted optimization procedure can be defined. A quantitative approach will be considered, preferably employing dedicated aircraft design decision making methods and techniques that can support complex decisions in multi-variable problems, also including lifecycle considerations and uncertainty analyses [16–20]. However, this

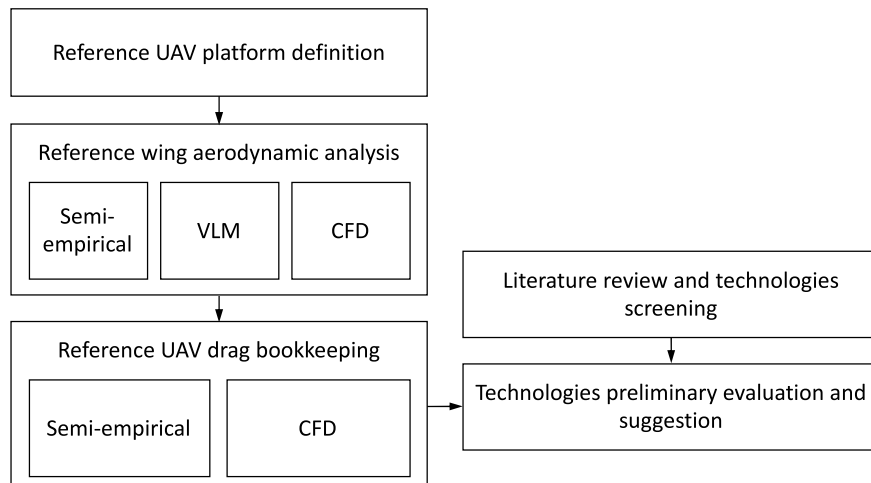


Fig. 2. Schematic description of the methodology.

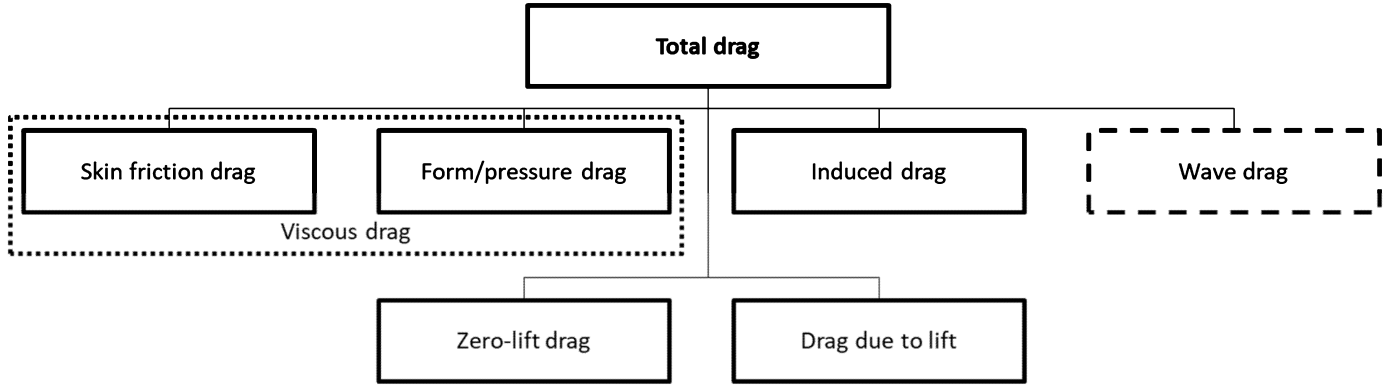


Fig. 3. Drag breakdown.

all-round approach is beyond the scope of the current work, which serves as the first step.

2.2. Drag force and drag bookkeeping

According to the fundamentals of aerodynamics, the drag force on a body can be broken down into four major components, depending on the underlying physics (Fig. 3, middle row). That is, the skin friction, the pressure, the induced and the wave drag components [21]. The sum of the skin friction and the form drag is called “profile drag” or “viscous drag”, whereas, in the case where an entire aerial vehicle is examined, the term “parasitic drag” is frequently used to describe the corresponding sum. The wave drag is only relevant in high-speed operations and further investigation is beyond the scope of this work, since the latter concerns fixed-wing UAVs that operate in low-speed, incompressible flow conditions.

Moreover, following a well-established and more practical approach for aeronautics, and excluding the wave drag term, another classification can be made, as presented in the bottom row of Fig. 3. Namely, the drag force at incompressible flow conditions can be broken down to the “Zero-lift drag” and “Drag due to lift”. The zero-lift drag ($C_{D,0}$) is the parasitic drag that exists when the lift of an aerial vehicle is zero. The drag due to lift depends on the lift coefficient and consists of the change in parasitic drag when an aerial vehicle produces lift, plus the induced drag from the wings and other lifting surfaces. The two components can be clearly identified in the well-known drag polar chart.

The drag breakdown analysis conducted in this study attempts to cover both of the above approaches, i.e. the physics and the practical, thus allowing for a more targeted discussion in the technology screening section.

2.3. Aerodynamic analysis tools

A combination of semi-empirical, low-fidelity, and high-fidelity aerodynamic analysis tools are employed in a synergetic approach. The semi-empirical calculations refer to textbook methods and, although being the fastest of the three, they are not very accurate when used on UAV configurations, as cited in [15]. The low-fidelity aerodynamic analysis methods refer to panel-based methods and are more time- and cost-efficient when compared to CFD. At the same time though, they have several disadvantages such as inability to accurately predict viscous effects, which in turn may result in underprediction of the drag force [22]. The high-fidelity methods refer to computational modeling, i.e. CFD. Apart from the relatively high computational requirements, the CFD results cannot yield the discretized contribution of each part, or the effect of each physics-related term (e.g. friction or separation). Hence, all

three methods are needed to increase prediction accuracy, as well as to distinct between the various sources of drag, both physics- and part-wise.

2.3.1. Semi-empirical methods

The theoretical aerodynamic relations and semi-empirical equations and charts of Roskam [23] are employed, which, as also mentioned in [15], are amongst the most accurate textbook methods. Its relations use the main geometric and aerodynamic parameters of each part as an input (e.g. the main wing mean chord, airfoil type, aspect ratio and taper ratio, the fuselage length and mean diameter etc.) and yield the lift and drag coefficients at the desired angles of attack and operational conditions. The drag component is estimated separately for each part and is then added to the total drag force. That is, for each component, the drag is broken down to its components, which are calculated using theoretical and semi-empirical relations. Such a relation is eq. (1), which is used to calculate the wing zero-lift drag coefficient:

$$C_{D_{0w}} = \frac{R_{wf} R_{LS} C_{fw} [1 + L'(t/c) + 100(t/c)^4] S_{wetw}}{S} \quad (1)$$

where the R_{wf} , R_{LS} , and R_{fw} are the wing/fuselage Reynolds number, the lifting surface correction factor, and the wing/fuselage friction coefficient, respectively. The rest of the parameters are related to thickness of the airfoil and the geometric characteristics of the wing, as explained in [23].

The lift component is estimated by summing the effect of the lifting surfaces. Indicatively, in the case where a wing and a horizontal stabilizer are used, eq. (2) is a relation derived from the theory of aerodynamics and can be used to estimate the lift coefficient as a function of the angle of attack α :

$$C_L = C_{L_0} + C_{L_a} \begin{cases} C_{L_0} = (i_w - \alpha_{0Lw}) C_{L_{\alpha wf}} \\ C_{L_a} = C_{L_{\alpha wf}} + C_{L_{\alpha h}} \eta_h \frac{S_h}{S} (1 - \frac{d_{\epsilon}}{d_{\alpha}}) \end{cases} \quad (2)$$

A detailed presentation of each term can be found in [23].

To facilitate the semi-empirical calculations, the Laboratory of Fluid Mechanics and Turbomachinery (LFMT) in-house tool is employed. Based on aircraft design methods, such as the ones described in [2,26,27], it also includes the aspects of unmanned aircraft design and is a combination of analytical and semi-empirical presizing methods. It can be used to carry out a complete layout design study and aerodynamic analysis and has been validated on commercial airliners and unmanned aerial vehicle configurations [21,28].

2.3.2. Low-fidelity methods

For the low-fidelity methods, the XFLR5 software was employed [24]. Based on the Xfoil airfoil analysis code, the XFLR5 can be

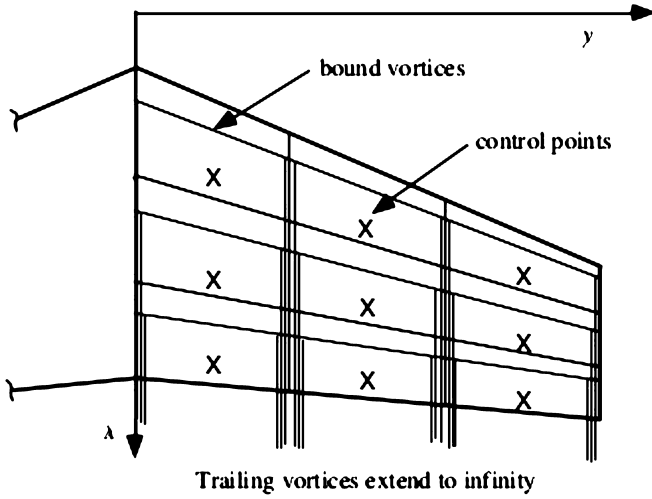


Fig. 4. Vortex Lattice Method panel arrangement for a plane [25].

used to conduct 2D and 3D aerodynamic analysis calculations around wings and bodies. It also supports direct and inverse airfoil design procedures (2D), as well as body design, center of gravity (CG) and inertia estimations, and stability calculations (3D).

For the 2D case, the analysis is based on the assumption that a general two-dimensional inviscid airfoil is constructed by the superposition of a freestream flow, a vortex sheet of strength γ on the airfoil surface, and a source sheet of strength σ on the airfoil surface and wake. The airfoil contour and wake trajectory are discretized into flat panels, with N panel nodes on the airfoil and N_w nodes on the wake. A panel of uniform source strength σ_{TE} and vortex strength γ_{TE} are also placed across the airfoil trailing edge gap if it has a finite thickness, whereas for smooth flow off the trailing edge the corresponding panel strengths σ_{TE} and γ_{TE} are related to the local airflow surface vorticity and the unit vector bisecting the trailing edge angle, and the unit vector along the trailing edge panel. By requiring that the stream-function is equal to some constant value Ψ_0 at each node on the airfoil, the following linear system is defined (eq. (3)):

$$\sum_{j=1}^N \alpha_{ij} \gamma_j - \Psi_0 = -u_\infty y_i + v_\infty x_i - \sum_{j=1}^{N+N_w-1} b_{ij} \sigma_j \quad (3)$$

where $u_\infty = q_\infty \cos \alpha$, $v_\infty = q_\infty \sin \alpha$ are the freestream velocity components and the coefficient matrices α_{ij} and b_{ij} are fully determined from the unit streamfunctions, if all the airfoil panel nodes x_i , y_i and the wake nodes are known. The linear system of eq. (6) is combined with a Kutta condition (eq. (4)):

$$\gamma_1 + \gamma_N = 0 \quad (4)$$

and gives a linear $(N+1) \times (N+1)$ system for the N node values γ_i and the airfoil surface stream-function. The above methodology is slightly adjusted for viscous flows, including the boundary layer equations.

In the 3D case, the analysis is conducted on a simplified representation of the wing/body configuration being investigated, using an inviscid approach. That is, the 3D configuration is modeled using a finite number of panels, and a vortex or a doublet and source is associated to each panel. Depending on the configuration, the analysis may be of the Vortex Lattice Method (VLM) type (Fig. 4), i.e. performed on the mean camber line, or of the 3D-panel type, in the case where the wing is modeled as a thin or a thick surface respectively [24]. More specifically, the classic Lifting Line Theory (LLT), a VLM, and a 3D Panel Method and their variants are

supported, whereas the boundary conditions may be either of the Neumann or Dirichlet type. A detailed presentation of their characteristics, their advantages, and their application limitations is made in [24].

2.3.3. High-fidelity methods (CFD)

The CFD analysis was conducted using the ANSYS CFX commercial software (ANSYS® Scientific Research, Release 18.1). The Reynolds-Averaged Navier-Stokes (RANS) equations are solved, coupled with the Spalart-Allmaras (S-A) turbulence model [25], to ensure high accuracy in the calculation of the normal and shear stresses acting on the wing's surface. The S-A turbulence model has been widely used in aeronautical applications and has been proven to be an adequate turbulence model for predicting turbulent or transitional external flows around airfoils and wings, since it is capable to correctly correlate the occurring strong pressure gradients with the developed boundary layers [26,27].

For a steady problem, the RANS equation in Einstein notation can be written as (eq. (5)):

$$\rho \bar{U}_j \frac{\partial \bar{U}_i}{\partial X_j} = \frac{\partial}{\partial X_j} \left[-\bar{p} \delta_{ij} + \mu \left(\frac{\partial \bar{U}_i}{\partial X_j} + \frac{\partial \bar{U}_j}{\partial X_i} \right) - \rho \overline{u'_i u'_j} \right] \quad (5)$$

In the Spalart-Allmaras turbulence model, the Reynolds stresses tensor is given by $-\overline{u'_i u'_j} = 2\nu_t S_{ij}$, ignoring the term containing the turbulence kinetic energy. S_{ij} is the strain-rate tensor and ν_t is the eddy viscosity, which is given by eq. (6):

$$\nu_t = \tilde{\nu} f_{\nu 1} \quad (6)$$

Note that $\tilde{\nu}$ is the "working variable" of the model, and obeys the following transport equation (eq. (7)):

$$\begin{aligned} \frac{D\tilde{\nu}}{Dt} = & c_{b1} [1 - f_{t2}] S \tilde{\nu} + \frac{1}{\sigma} [\nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu}) + c_{b2} (\nabla \tilde{\nu})^2] \\ & - \left[c_{w1} f_w - \frac{c_{b1}}{\kappa^2} f_{t2} \right] \left[\frac{\tilde{\nu}}{d} \right]^2 + f_{t1} \Delta U^2 \end{aligned} \quad (7)$$

Details about the variables and the various coefficients can be found in [25].

A second order upwind scheme is used for both the discretized momentum and turbulence model equations, whereas, concerning the inlet turbulence boundary conditions, the turbulence intensity is set at 1% and the eddy viscosity ratio at 0.2 [28]. A steady-state approach is employed due to computational resources limitations. An unstructured grid is generated around each of the examined configurations, whereas several inflation layers were generated on the walls of the examined configuration to ensure that y^+ at the wall will not exceed a value of 5, so that the local, low-Reynolds boundary layer phenomena in the wall region can be accurately modelled. For each of the employed grids, dedicated grid dependency analysis was conducted, using the lift and drag coefficients as the monitored variables, as indicatively presented in Fig. 5. Ultimately, that lead to the selection of approximately 6,000,000 nodes for the main wing and 9,000,000 nodes for the integrated UAV configuration. Note that those grids include 20 inflation layers around the wing external surfaces, the first of which has a thickness of 2.5×10^{-5} m, so that y^+ does not exceed the value of 5 at any point on the wing surface.

3. The reference UAV configuration

The HCUAV RX-1 serves as the reference configuration of this study. Its aerodynamic design has been conducted by LFMT as a part of the HCUAV project, whose aim was the development of a MALE UAV prototype for civil operations in Greece (Fig. 6). The first prototype, marked as RX-1, made its maiden flight in the summer

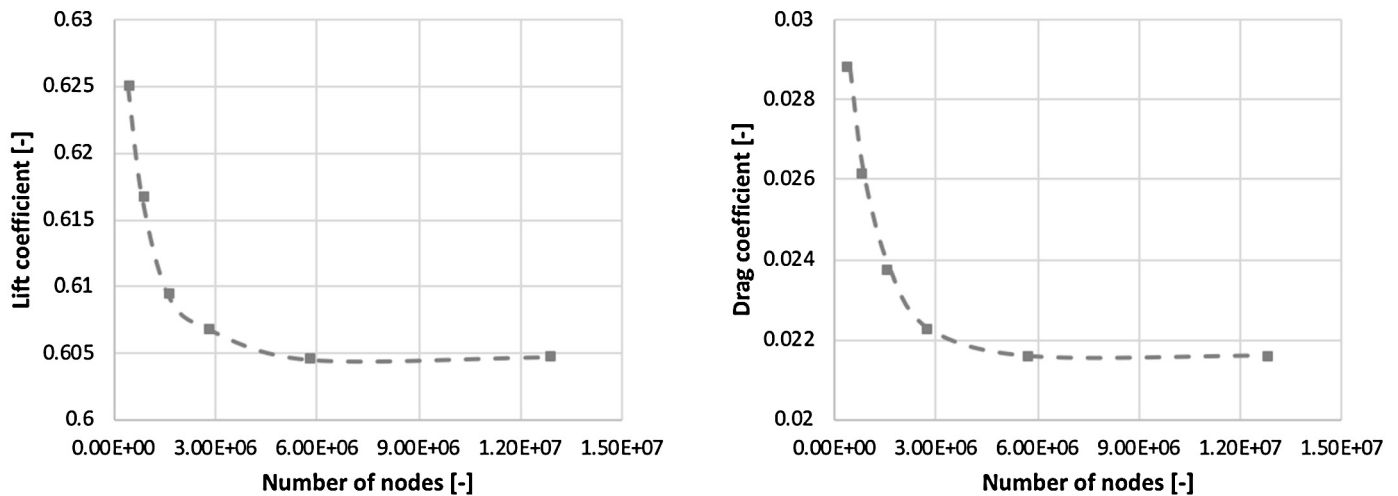


Fig. 5. Grid dependency study for the HCUAV RX-1 main wing at an angle of attack of 2 degrees.

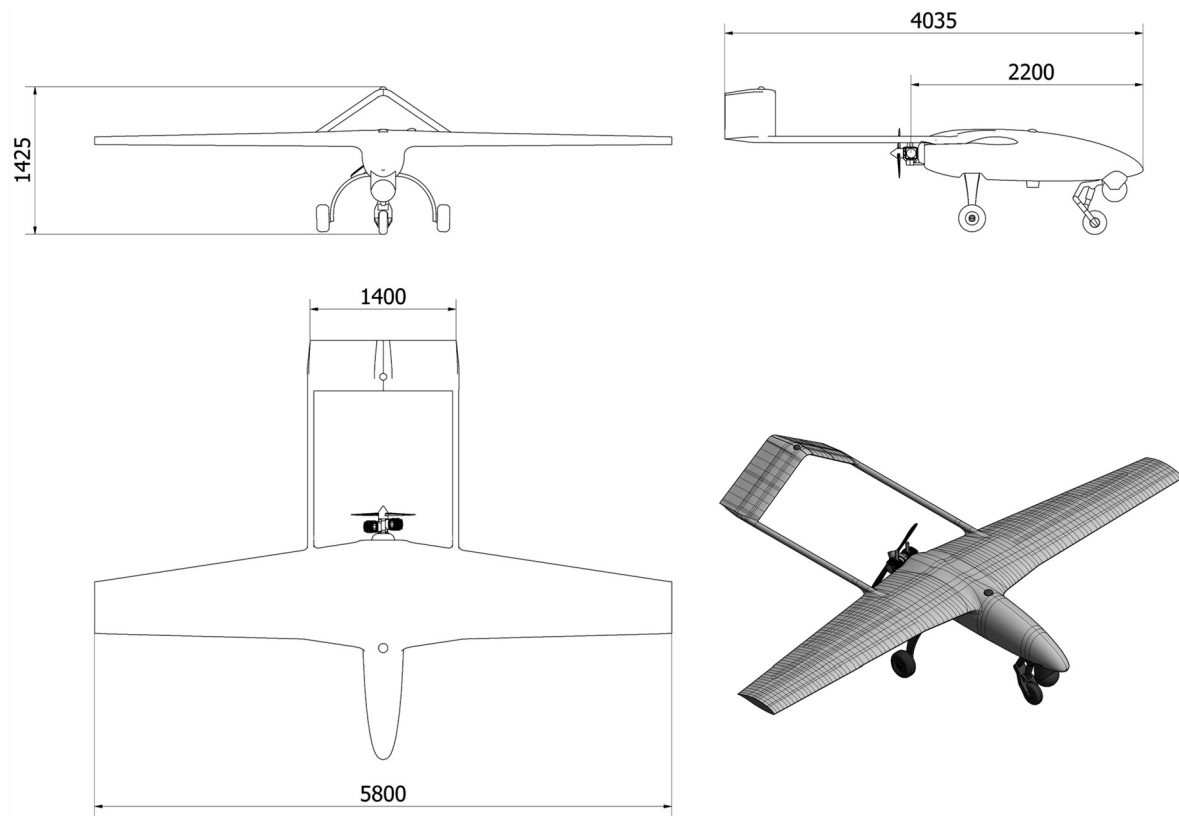


Fig. 6. The HCUAV RX-1 3D CAD before its optimization.

of 2016. Resembling a rather typical fixed-wing UAV configuration, it is selected as a reference point for the current study, since the authors have all the information (e.g. configuration layout, operating conditions etc.) required for a thorough aerodynamic analysis investigation, whereas lots of this information, along with the design procedure details, are available in the literature [5,29,30]. In the following paragraphs, a brief presentation of the basic configuration specifications of every part is made, emphasizing on its most important characteristics and design philosophy. More information about the mission requirements, the design procedures, and the optimization loops can be found in the corresponding papers.

The specifications of the wing platform are summed up in Table 1. A thick, high-lift, Natural Laminar Flow (NLF) airfoil [31]

is used. It combines excellent aerodynamic performance characteristics (high L/D ratios, lift-curve slope, stall angle) for the corresponding Reynolds numbers (1.9×10^6 , based on the mean aerodynamic chord) with large internal space. Moreover, the wing has an Aspect Ratio of 7.45, and a taper ratio of 0.45, which are amongst the typical values for MALE UAV wings [32–34].

The fuselage is designed to have enough room for the electro-optical equipment to be placed and for the engine to be installed, and at the same time, ensure their proper cooling. To keep the drag at a minimum, the fuselage is streamlined using airfoil profiles, whereas dedicated analyses were made to appropriately position the engine, size the cooling ducts, as well as to integrate the wings and landing gear. Finally, the empennage has an inverted-V shape, appropriately sized to satisfy the stability and control requirements

Table 1
HCUAV RX-1 wing specifications.

Wingspan	5.6	m
Centerline chord	1.1	m
Kink chord	0.85	m
Tip chord	0.5	m
Taper ratio	0.455	[-]
Aspect ratio	7.45	[-]
Airfoil profile	NFL 1015	

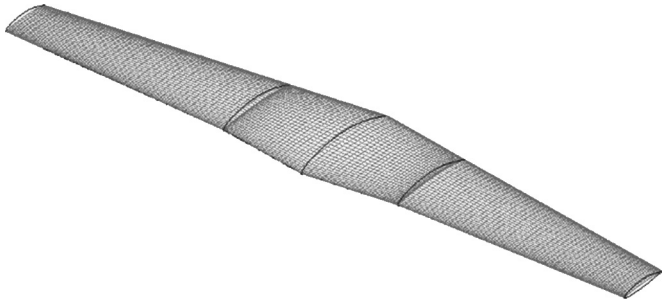


Fig. 7. Panel arrangement over the reference wing for the low-fidelity aerodynamic analysis.

of the HCUAV RX-1. It is mounted on the main wing using a pair of booms. The remaining parts are the camera gimbal and the landing gear. Concerning the latter, a non-retractable configuration was selected, due to the lack of internal volume and to avoid weight and complexity penalty.

4. Results

4.1. Reference wing aerodynamic analysis

For the aerodynamic analysis of the reference wing all the available methodologies are combined, to compare the results and validate the tools on this relatively simple configuration. Regarding the semi-empirical aerodynamic analysis, the examined configuration is a conventional one, therefore no modification is made to the traditional methods of [23]. Regarding the low-fidelity method, a VLM approach is selected, since the examined configuration is a planar geometry, with a very small sweep angle and a medium

aspect ratio. A total of 80 and 120 panels are generated on the streamwise and spanwise directions, respectively (Fig. 7), since increasing their number has little to no effect on the generated lift and drag coefficients. Concerning the CFD analysis, a mesh of approximately 6,000,000 computational nodes is generated around the reference wing configuration (Fig. 8), following the grid dependency study described in Section 2.3. The inlet conditions correspond to the reference conditions, i.e. a freestream velocity of 140 km/h and a flight altitude of 2000 m [5]. A Reynolds number equal to 1.9×10^6 is calculated, based on the mean aerodynamic chord of the wing.

The results of the aerodynamic investigations are presented in Fig. 9, whereas Fig. 10 shows the flow development at the tip of the wing as a result of the computational simulations. Considering the lift curve slope, the percentage error between all three methods does not exceed 5% up to an angle of attack of 8° . At larger angles of attack though, the low-fidelity and the semi-empirical curves diverge from the CFD results, and that is attributed to the fact that these methods cannot accurately predict near- and post-stalling behavior due to viscous effects. When comparing the drag polar results, the semi-empirical and low-fidelity curves almost coincide with each other, but the ones derived from the computational simulations have an offset of approximately 30–50 counts of drag, or 10%, to the bottom-right of the chart. That is, the lower fidelity methods underpredict the parasitic drag term, an observation that is in agreement with previous studies as well [22,24].

Based on the results of the aerodynamic analysis, a breakdown of the drag term is made for the wing configuration. The computational simulations may yield the most accurate results between the three aerodynamic analysis methods, however the only distinction that can be made is splitting the total drag force into the zero lift drag and the drag due to lift terms (Fig. 11, left). As is evident, the drag due to lift is small in negative angles of attack but becomes prominent from 4° onwards. Note that, at -4° the drag due to lift is negative, a common observation in wings made up from cambered airfoils. The VLM results on the other hand, can provide the decomposition of drag into its physical components, i.e. the induced and the viscous drag. As, in terms of lift, the VLM and CFD results are in very close agreement (Fig. 9), it can be assumed that the VLM accurately predicts the lift distribution over the wing configuration. Hence, it can also be assumed that the corresponding inviscid drag component, i.e. the induced drag, is also

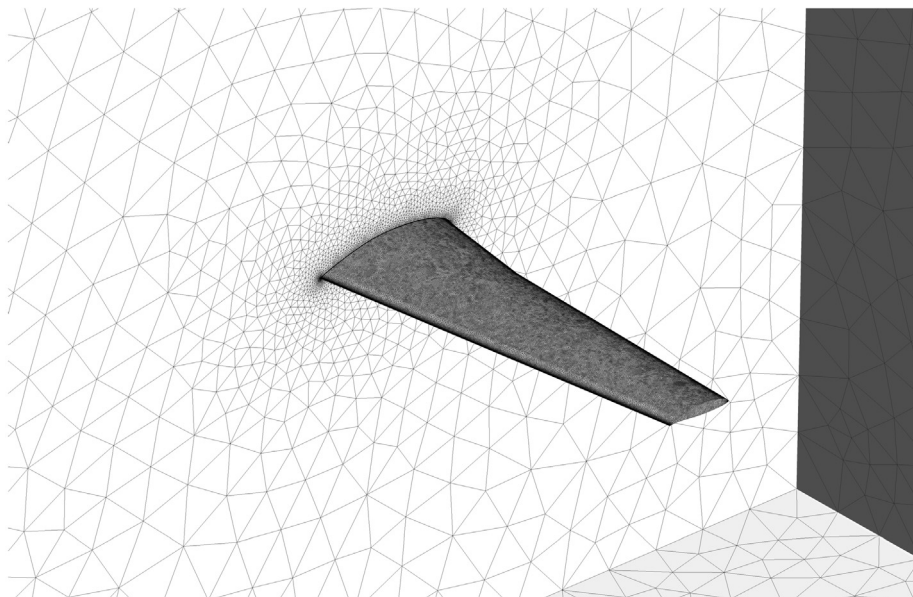


Fig. 8. Computational domain around the HCUAV RX-1 wing.

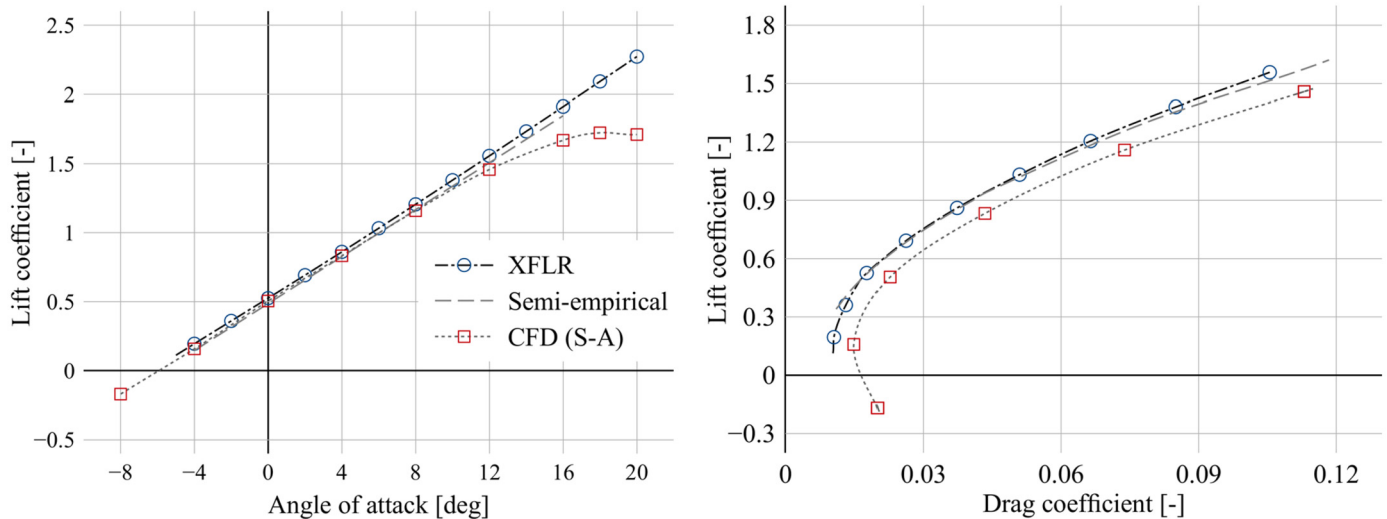


Fig. 9. Lift curve vs angle of attack (left) and drag polar (right) investigations for the reference wing configuration.

accurately predicted by the VLM, and that the deviation in the drag polar can be attributed to an underprediction in viscous drag. The corresponding result is presented in the right-hand side of Fig. 11.

4.2. UAV drag bookkeeping

The wing is the most important part of a UAV, and of a fixed-wing aerial vehicle in general, when it comes to lift production. However, several other components contribute to the drag force. To quantify this contribution, an aerodynamic analysis is conducted for the entire HCUAV RX-1 configuration. A combination of the in-house aerodynamic analysis tool and CFD are employed, since the low-fidelity aerodynamic analysis methods cannot model a complex 3D configuration and accurately predict the viscous drag produced by its parts [24].

The HCUAV RX-1 lift coefficient vs angle of attack and drag polar are presented in Fig. 12. In a similar fashion to Fig. 9, the two methods are in close agreement up to an angle of attack of 8 degrees. From this point onwards, the lift curves deviate, since extensive viscous separation occurs over the wing. Once again, the CFD investigations can be trusted for their accuracy but cannot provide any insight regarding the sources of the generated drag force. On the contrary, the semi-empirical methods may not be as accurate, but they calculate the drag force for each part separately. Based on the aforementioned results, Fig. 13 presents the drag force generated by each part as a percentage of the total drag coefficient (100%) for the angles of attack that correspond to the majority of the mission (cruise and loiter flight segments). To support this selection, and although providing more insight about the performance of the aerial vehicle is beyond the scope of this study, an indicative value for the mid-loiter lift coefficient is 0.528 [29]. As is evident, the main wing is the main contributor of drag, with the empennage and fuselage ranking second, and third, respectively. The landing gear and miscellaneous drag components (camera gimbal, cooling drag, etc.) have a smaller, yet considerable, contribution to the total drag budget.

An important remark must be made at this point. The results of the drag bookkeeping analysis will vary between different fixed-wing UAV configurations. However, by comparing the results of the current work with the ones of [15], and even though their drag analysis results are mostly limited on zero-lift drag coefficients, the drag breakdown between the medium-sized UAVs examined there (wingspan between 4 and 9 meters) and the one examined in the current study, present very similar distributions. Indicatively, the predicted percentages for the wing, fuselage and empennage

present deviations around 5-10% for the zero-lift case. The authors consider this difference acceptably small, given the fact that different UAV configurations and modeling methods are employed between the two studies. Hence, Fig. 13 can serve as a reference point for aerial vehicles that do not have any radical changes in the design philosophy (e.g. flying wings) and operate in similar conditions and Reynolds numbers, since the main trends will not deviate drastically.

4.3. Technology screening

In this section, a presentation of technologies and techniques for drag reduction and aerodynamic and performance enhancement is made. Most, if not all, of the options listed here have been investigated, developed, or proposed for larger-scale manned aerial vehicles and jet transports, as also presented in [35,36]. Although the list may not be exhaustive, an attempt to broadly cover the most important technologies and techniques is made, whereas the unique aspects of UAVs, such as the drag bookkeeping information, operating conditions, weight, complexity, and cost limitations, are considered as well.

4.3.1. Conventional layout options

As a start, a considerable reduction in structural weight, and a subsequent enhancement in performance, can be achieved through the utilization of Carbon Fiber Reinforced Polymers (CFRPs). The structural weight reduction has a direct impact on the empty weight and, consequently, the Gross Takeoff Weight (GTOW) of a fixed-wing UAV configuration, positively affecting all performance specifications. More specifically, according to the weight sizing methods of [23], the use of CFRPs results in a reduction of the structural weight of the wing, empennage, fuselage, and landing gear. That can indicatively be seen in eq. (8), which is proposed for the wing configuration of General Aviation aircraft, and is also suitable for estimating the weight of a UAV wing [37]. The factor F_c is applied to count the effect of the material during the presizing calculations. When CFRPs are used, the factor F_c is set at 0.85, for a 15% reduction. Note the methodology is similar for the rest of the parts. One possible drawback for the use of CFRPs when compared to traditional alloys, is the lack of analytical sizing methods, computational modeling techniques, and layout examples available in the literature, which means that a layout designer will need a more meticulous approach and, preferably, in-house experimental testing to achieve an optimum structural layout.

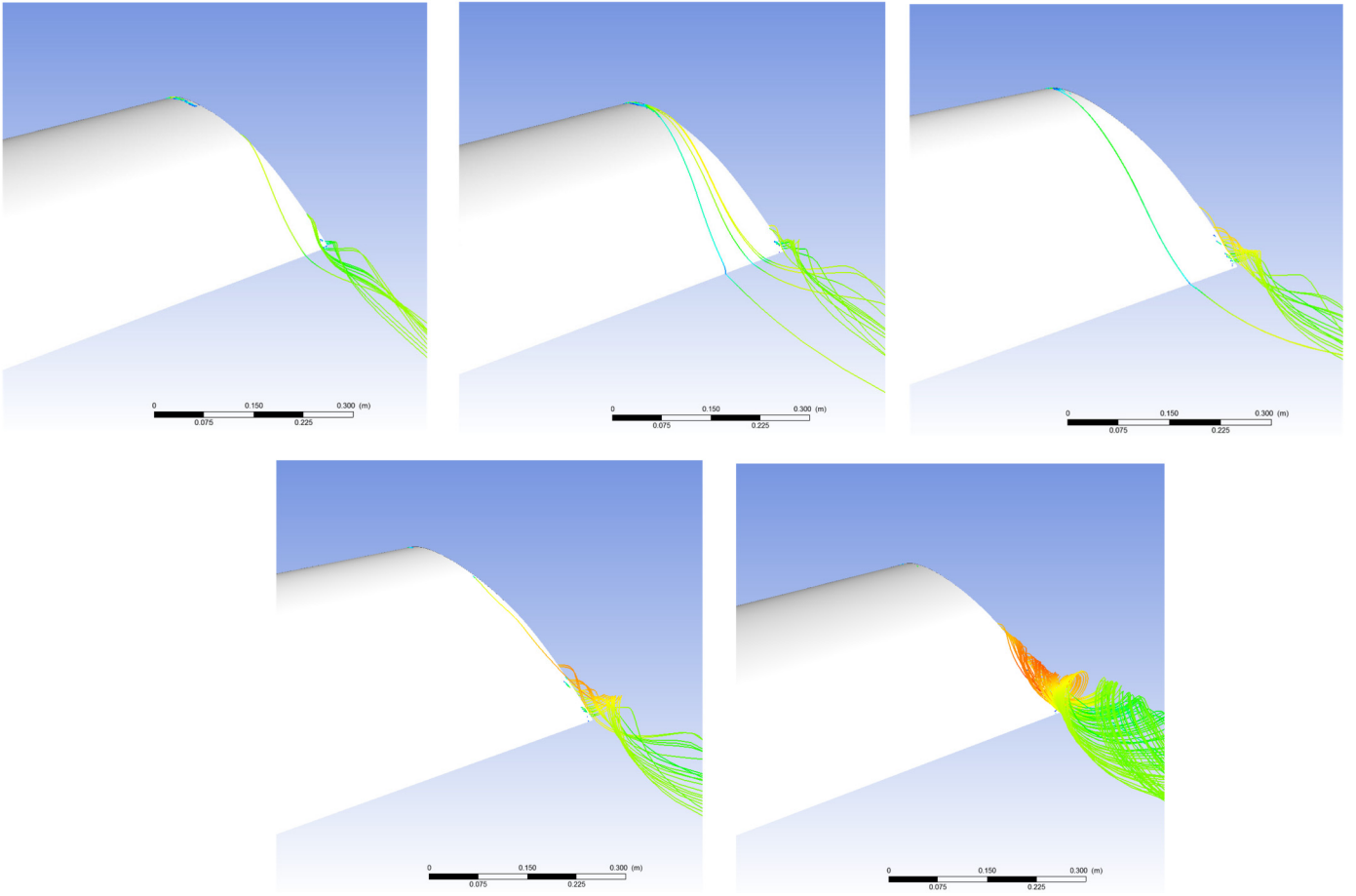


Fig. 10. Tip vortex formation at the reference wing platform.

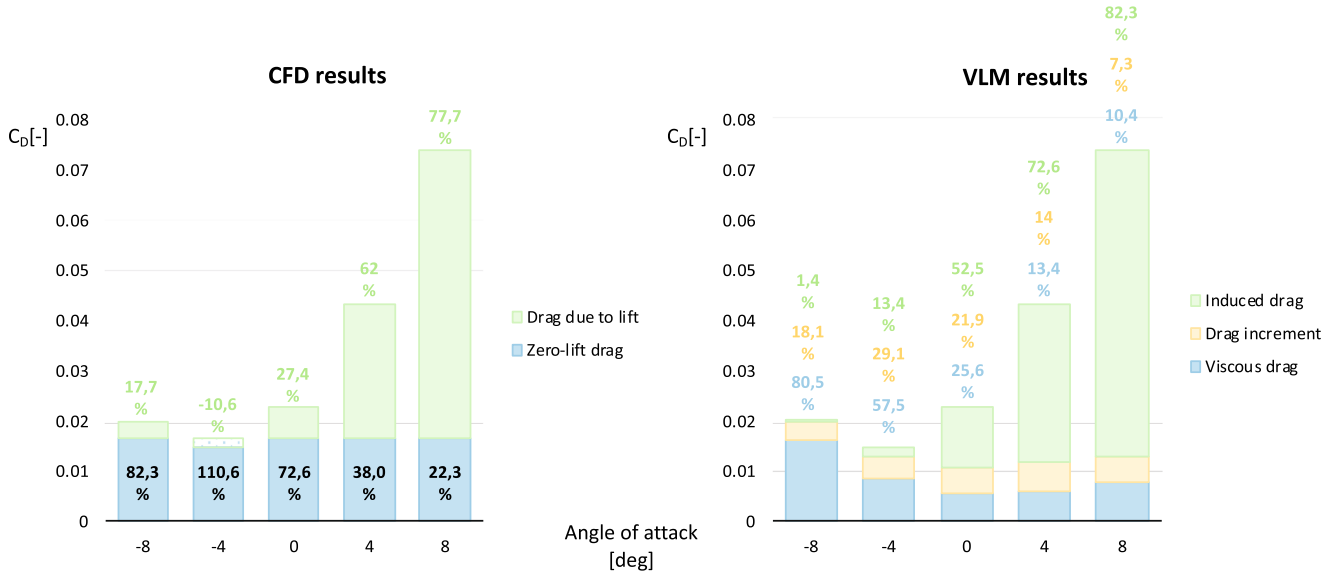


Fig. 11. Drag breakdown for the reference wing configuration, based on the aerodynamic investigations.

$$W_{(wing)} = F_c F_w 0.036 S_w^{0.758} W_{fw}^{0.0035} \left(\frac{A}{\cos^2 \Lambda} \right)^{0.6} q^{0.006} \times \lambda^{0.04} \left(\frac{100t/c}{\cos \Lambda} \right)^{-0.3} (N_z W_{dg})^{0.49} \quad (8)$$

Concerning aerodynamics, there are several guidelines and methods from traditional aircraft sizing, which can be utilized by

a UAV designer as well. Regarding the wing, which generates the majority of the drag, there are several layout parameters that affect drag production. As a start, the airfoil selection plays an important role as, according to the operational Reynolds numbers, the airfoil profile has a considerable impact on profile drag. The use of natural-laminar airfoils can be considered, to increase the laminar region over the wing (reaching as much as 60% in chord length),

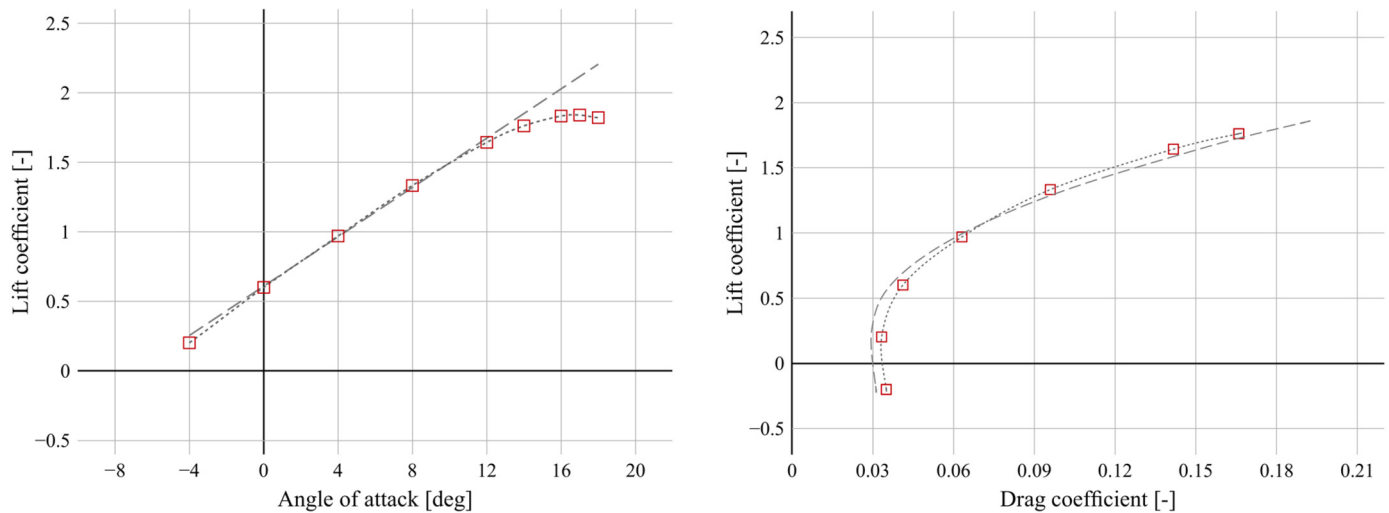


Fig. 12. Lift coefficient vs. angle of attack (left) and drag polar (right) for the HCUAV RX-1 reference configuration. The dashed line presents the results of the semi-empirical method, and the dotted line with the red markers the CFD ones.

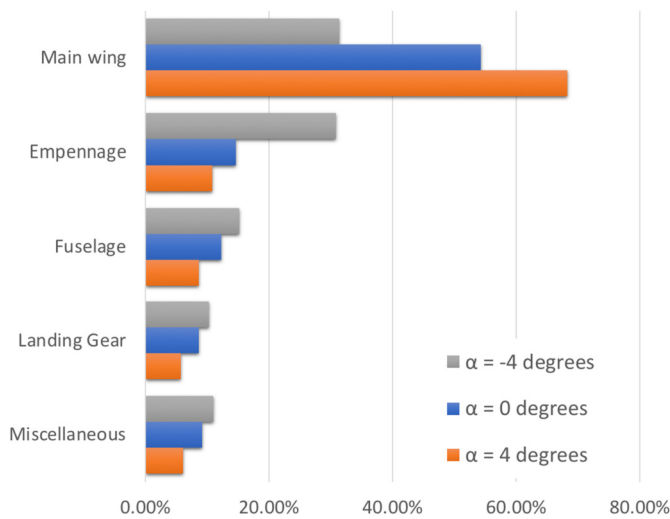


Fig. 13. Drag force breakdown between the HCUAV RX-1 components using the semi-empirical aerodynamic analysis tool for the examined angles of attack.

thus reducing friction drag term [38] that is dominant at low angles of attack. When it comes to the 3D layout, the reference area is usually selected based on wing loading and performance requirements. The well-known parameters of aspect and taper ratios can be fine-tuned, using optimization methods, since they have a considerable effect on aerodynamics and the induced drag term, which has a considerable effect at operating angles of attack [21]. Both variables also have an impact on structural weight as well, therefore their selection must be carefully made, based on presizing calculations, mission type trade studies, and common practice.

The empennage part is mainly responsible for the stability of the aerial vehicle, hence its size is more-or-less defined from the corresponding calculations. However, with its contribution to the total drag budget ranging from 10% to 30%, the empennage drag must not be overlooked at. Trade studies between the size and the longitudinal location can aid in achieving an optimal weight/aerodynamics combination, whereas the shape can also affect the overall performance [37]. For example, if properly sized, the ruddervators (combine the function of the elevator and rudder control surfaces) of a V-tail configuration reduce the need of control surfaces from three to two, compared to a conventional tail arrangement.

The required actuators and systems are in turn reduced from three to two as well. Hence, a slight, yet considerable, weight reduction is achieved at the far rear of the aerial vehicle, leading to a reduced need for ballast weight at the front.

Concerning the fuselage, the options are relatively limited, since its shape and size are mainly defined based on the needs for operability and maintainability. Given that the above constraints are met, though, the fuselage shape can be streamlined to reduce the generated drag force, which roughly contributes approximately 10% to 15% to the total drag force. Streamlining usually comes at the cost of more complex shapes, but this issue can be overcome by the modern composite materials technology and molding manufacturing techniques.

Another layout option to enhance the aerodynamic efficiency of an aerial vehicle is the winglet configuration. It is extensively used in aircraft design, from large commercial airliners to small sailplanes, to perturb the effect of the wingtip vortex and reduce the induced drag component, thus yielding considerable improvements in aerodynamic efficiency [36,39,40]. Especially in UAV development, where the time and budget are limited when compared to a commercial airliner, the wings are of less complex geometry (e.g. in terms of aerodynamic and geometric twist) and simpler to manufacture. This in turn means that the loading at the wingtips of a UAV is usually higher. Moreover, to ensure small transport and deployment costs, the wingspan is also limited. Combining the above with the relatively low operating speeds, means that the induced drag component plays a dominant role in a UAV wing, which in turn generates the largest amount of drag. The latter can be evaluated by the results of the previous section. That is, the induced drag contributes as high as 50% to the drag force of the main wing, which in turns produces 53% of the total drag force at an angle of attack of 0 degrees. The corresponding percentages become 70% (induced term contribution to the wing drag) and 67% (main wing contribution to the total budget) at 4 degrees. Therefore, the winglet configuration appears to be an ideal match for fixed-wing UAV aerodynamics, and at the same time is a low-complexity, low-weight, low-cost solution. However, as also cited in [14,40], due to the complex nature of the tip vortex phenomenon, an optimized winglet's geometry will vary, depending on the size of the aircraft and its mission. A bad design might well decrease performance characteristics, hence a winglet configuration must be carefully designed and tested before installed on a wing. The positive effects of well-designed winglets on UAV performance is indicated by recent research [14,22,41], where a greater-than-10% total drag reduction

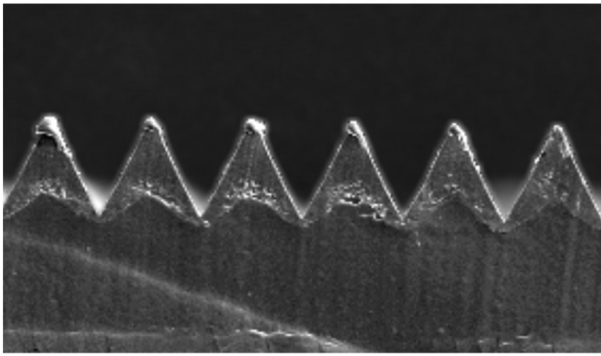


Fig. 14. Riblet film viewed under a scanning electron microscope [42].

for on-design conditions is reported, given that dedicated, high-fidelity winglet optimization methods are employed.

4.3.2. Flow control techniques

Flow control technologies can generally be divided into passive and active, and are mainly used to contain separation and delay stall, or to delay the laminar-to-turbulent transition. Their main difference is that the active systems require a power input for their operation, whereas the passive ones do not. A typical example of passive flow control technique is the vortex generators. They are made up of bumps or small aspect ratio airfoils, mounted normally to the lifting surfaces, ahead of the flow separation point to energize the boundary layer and to prevent separation. However, this comes at a considerable cost in drag, therefore no further analysis will be made [36,42].

Another way of controlling the flow without the need for power input, is by applying a coating on the external surface the aerial vehicle, which interacts with the overlying flow and reduces friction drag. One of the few techniques that have been successfully applied to the reduction of the skin friction in turbulent boundary layers, both in laboratory conditions and in small-scale aerodynamic configurations, is the riblets coating [43]. Riblets are essentially small surface protrusions (Fig. 14) aligned with the direction of flow, typically in the scale of 40–150 μm , and have been investigated by various researchers, mainly for aeronautical applications [42–47]. The typical documented gains in drag reduction are in the order of magnitude of 5–10% for flat plates and wing configurations. The percentage sometimes refers to the reduction of either viscous or total drag [44], and the effect of the riblets greatly depends on the relation between their geometrical specifications and operational Reynolds number, which makes it difficult for a clear and definitive verdict to be made. The above observations, along with their relatively low Technology Readiness Level (TRL) and the fact that an aging process deteriorates their performance over time [48], rules out riblets as a viable option for the time being. However, this by no means that the technique is to be dropped entirely for UAV applications. Based on the results, the rapidly advancing manufacturing techniques [49,50], and the ongoing aerodynamic research, the riblets could be established as a passive flow control technique in the near future.

Regarding the Active Flow Control (AFC) techniques, perhaps the most commonly encountered one, is using high-lift devices at the wings, i.e. trailing- and leading-edge flaps, or slats. They can be used to enhance the lift production and maximum lift coefficient of a wing configuration, or to delay stalling, thus greatly relaxing the need for low wing loading at takeoff, landing, and approach conditions. High-lift devices though, have a significant negative impact on aerodynamic efficiency and stability upon deployment [35,36], whereas they come at a considerable weight penalty and system complexity. Nevertheless, the methods for sizing and investigating the effect of those devices are well-established [51], and

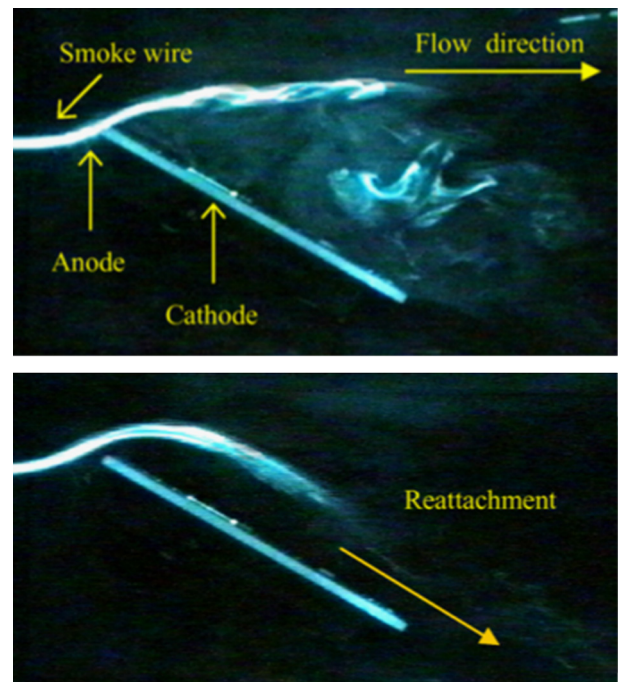


Fig. 15. Separation delay using the DBD PA technique [62].

can be used to conduct dedicated trade studies, if considered appropriate.

Another, more sophisticated, AFC technique is the Variable Camber system (VC). It employs trailing edge flaps to change the pressure distribution of the airfoil/wing [35]. Optimum aerodynamic efficiency can be achieved, especially for off-design conditions, leading to a corresponding increase in L/D in cruise of approximately 2–3% [52]. Similar morphing wing and airfoil technologies are documented in detail in [53,54]. The idea of changing the incidence and twist angles of a wing, or the camber of an airfoil mid-flight, offers several advantages in aerodynamic efficiency of an aerial vehicle, in general. However, those technologies come at a very large weight penalty, and cannot be considered as a viable solution for medium-sized UAVs yet.

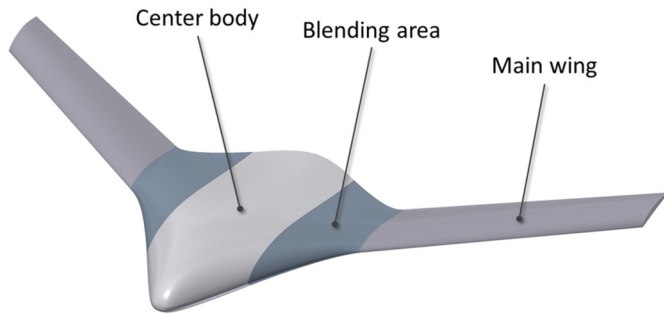
A large category of AFC techniques involves the use of air blowing or suction actuators, of various types [36,55]. However, the majority of those actuators have been developed for larger-scale aircraft and require extensive modifications on the external surfaces and/or its internal parts, such as heavy and complex piping systems, mechanical devices and air control systems, and holes or slots, which are prone to contamination, dust or insect ingestion, to name a few. Moreover, there is often a need for mass flow supply, which in the case of a jet transport can be covered by the turbofan engine bleeds. All of the above requirements pose serious feasibility issues for the smaller-scale UAVs.

On the other hand, those issues do not apply in the case of the Dielectric Barrier Discharge (DBD) plasma actuator (PA). It is an innovative active flow control technique, based on the formation of a low-temperature plasma between a pair of asymmetric electrodes that are separated by a dielectric material (Fig. 15). As a result, an ionic wind, similar to a wall-jet, is generated and momentum is added into the boundary layer. The DBD PA technique has been under investigation in many studies for the last two decades [56–62]. Indicatively, a potential increase in maximum lift coefficient of about 15% [62] has been documented at a Reynolds number of 2.25×10^6 and $M = 0.3$. The flow conditions correspond to the ones encountered at tactical and MALE UAVs, whereas the integration cost is 100 W and 1.36 kg (including the required generator) per meter of actuator length [56], totally

Table 2

Technology screening for fixed-wing UAV applications.

Proposed technology/technique	TRL	Potential benefit(s)	Feasibility issue(s)	Suggestion
CFRPs	9	Weight reduction	None	●
Winglets	9	Aerodynamic efficiency/performance enhancement	Lack of high-fidelity design methods	●
VGs	9	Stall delay	Drag penalty	●
Riblets	4	Stall delay/drag reduction	Low TRL	●
High-lift devices	9	Lift enhancement/stall delay	Weight penalty/integration complexity	●
Morphing airfoils/wings	4	Lift enhancement/stall delay/drag reduction	Weight penalty/integration complexity/Low TRL	●
Blowing/suction AFC actuators	7	Stall delay/drag reduction	Weight penalty/integration complexity	●
DBD PAs	4	Stall delay/drag reduction	Low TRL	●
BWB	6	Aerodynamic efficiency/performance enhancement	Lack of design methods	●

**Fig. 16.** The BWB configuration.

acceptable for that class of aerial vehicles. Regarding the drag reduction potential, a value of 5% has been calculated for the wing configuration, at low-speed conditions [63], which is relatively low. The main issue that prevents the extensive use of the DBD PA technique though, is that their performance at various flight altitudes, weather conditions, and contamination levels has not been widely investigated. In lab conditions it is an energy efficient technique with very rapid response times and, combined with the fact that DBD PAs are lightweight devices with no moving mechanical parts or any need for air bleeds, it is evident that they have a considerable advantage over other active flow control techniques, at least as far as UAV applications are concerned. Hence, further research must be conducted to reach a safe conclusion regarding this technology.

4.3.3. Novel configurations

Several novel aircraft configurations have been proposed in the last decades, as potential candidates for short- and long-range commercial airliners. Advanced, high aspect ratio tube-and-wing [35], strut braced [64] configurations, as well as non-planar, boxed- or joint-wing wing aircraft [65,66] are amongst those configurations. Arguably though, the most revolutionary, in terms of shape, and the most promising, in terms of efficiency, is the Blended Wing Body (BWB) configuration (see Fig. 16) [67]. Indicatively, preliminary studies mention a 20% improvement in aerodynamic efficiency and a 27% reduction in fuel burn per passenger [67–69]. Due to issues related to cabin pressurization, passenger safety and evacuation [70–73], this novel configuration cannot be considered as a viable option for a commercial airliner platform before the late 2030s [70]. However, in the case of a UAV, where there is no human on-board, the BWB can be fully exploited. More specifically, recent studies [13,74] point out that there is a considerable benefit from the use of the BWB configuration for UAV applications with the two most important advantages being the enhanced aerodynamic efficiency and increased internal volume.

4.3.4. Overview

A wide range of possible technologies and techniques for aerodynamic efficiency and performance enhancement has been listed and evaluated for UAV applications, as summed up in Table 2.

The potential benefits, TRL and feasibility issues are briefly stated for each technology/technique. In the last column, the authors suggestion is shown, using a simple three-color scale: green for “applicable”, orange for “research/trade studies needed”, and red for “dropped”. Following this analysis, it follows that CFRPs, the winglet configuration and the BWB configuration are the most prominent solutions, since they combine a high potential with a satisfactory readiness level. Research on some of the other techniques, such as the riblets and plasma actuators, is also highly advised.

5. Conclusions

In the present work, a step-by-step investigation of aerodynamic efficiency and performance enhancement methods and techniques is made for fixed-wing UAVs that operate in the low-speed, subsonic regime. As a start, an aerodynamic analysis and drag breakdown is made on a reference UAV wing platform using a combination of semi-empirical textbook methods, low-fidelity and high-fidelity aerodynamic coefficient estimation tools. Then, a drag bookkeeping study is made for the entire UAV configuration, presenting the contribution of each part to the total drag budget for a range of operational angles of attack. Using the results of the aerodynamic investigations and the unique UAV specifications as a guide, a thorough drag-reduction technologies screening is made based on existing literature. Their potential advantages, as well as integration and realization issues are discussed. Concluding, the most promising technologies are proposed taking into account the potential benefits as well as the technology readiness.

Regarding the employed methods, it can be stated that no aerodynamic coefficient estimation method can accurately predict all the required information when it comes to drag prediction for fixed-wing UAVs. The combined use of methods or, better yet, a calibration of the existing textbook methods is suggested, adjusted to the configuration and operating conditions of UAVs. Moreover, the drag bookkeeping data for tactical and MALE UAVs available in the literature are very limited in terms of examined configurations and operating conditions, even though those are the largest categories in terms of market share, in the already booming UAV market. The results of the current study are in agreement with the already existing ones and can be used as a reference case for future studies.

Concerning the results of the technology screening investigation it can be concluded that most, if not all, of the drag reduction technologies for commercial airliners and general aviation aircraft can be considered for UAV applications, provided that some special considerations are taken into account. For example, the fact that unmanned vehicle development has limitations in terms of budget and time, means that the use of some techniques for UAVs is either discouraged (e.g. high-lift devices) or encouraged (e.g. winglets). The different operating velocities and corresponding Reynolds numbers can also work in favor of some techniques, as is the case with the DBD PAs. On the other hand, the extra integration/installation weight for some of the performance enhancing

techniques rules them out as a viable option for UAVs, due to their smaller overall size and weight of those aerial vehicles. Finally, the fact that UAVs have no pilot or human being on board, means that lots of the airworthiness regulations do not apply and that more disruptive configurations can be exploited. Such is the case of the BWB, where the absence of humans on-board means that most of the configuration disadvantages no longer apply.

To sum up, and even though the list may not be exhaustive, this work provides some initial guidelines and suggestions for UAV aerodynamic efficiency and performance enhancement. The methodology of this study can also be applied to other fixed-wing aerial vehicles, ranging from both smaller and larger UAVs, to light and ultra-light airplanes and to cargo aircraft, provided that the operating conditions, Reynolds numbers and design philosophy is similar.

Regarding future work, the development of more accurate low-fidelity aerodynamic analysis methods could aid in a more detailed drag bookkeeping analysis and, in turn, a more targeted investigation. Moreover, when it comes to the techniques themselves, some of the most promising ones are in very low TRL for a definite conclusion to be drawn. For example, more research is required on morphing configurations and DBD PAs, before an essential evaluation for UAVs applications can be conducted. Most importantly though, it should be noted that to go beyond the suggestions of the current study and reach a solid conclusion concerning the technologies selection, other aspects such as costs and compatibility with other technologies should also be considered. This would in turn require a more detailed and quantitative analysis, preferably by employing dedicated decision support methods. Such a detailed analysis may be beyond the objectives of this initial study, however it is highly advised for future research to draw more solid guidelines for UAV aerodynamic efficiency and performance enhancement technologies.

Declaration of competing interest

The authors declare that no conflict of interest exists.

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